

GUIDELINES/CHECKLIST FOR PROJECT DOCUMENTATION

Before final submission, each FYP advisor and his students are required to ensure the following check list in the FYP documentation. Some examples of below mentioned clauses are also provided for your guidance.

Clause 1: **Table of Contents** shall be generated using MS Word Table of Contents method. (Manual numbers should not be accepted). Chapter one shall start from number 1 and before chapter one all number should be roman. Table of content will be of same color and same sized font, as shown **FIGURE 1**.

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FIGURE 1. Sample example of Clause 1.

Clause 2: All **Tables** shall be captioned (using proper MS Word insert caption method on top of table), explained in detail and referred in Text. Every table shall be captioned with chapter number

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FIGURE 2. Sample example of Clause 2.

serial wise. For example, first table in chapter one captioned as 1.1 and second table in chapter one as 1.2 and so on. Similarly, first table in chapter two captioned as 2.1 and second table in chapter two as 2.2 and so on. Captions should be consistently aligned (preferably centered). Elaborate each table appropriately, with at least 3 lines of description. Tables should not be split on two or more pages. In case of a large table that could not be fit on a single page, put header row on top of each page's split part of table.

Clause 3: **List of table** shall be generated using MS Word Table of figures/table method.

Clause 4: All **figures** shall be captioned (using proper MS Word insert caption method on bottom of figure), explained in detail and referred in Text (see **FIGURE 2**). Every figure shall be captioned with chapter number serial wise. For example, first figure in chapter one captioned as 1.1 and second figure in chapter one as 1.2 and so on. Similarly, first figure in chapter two captioned as 2.1 and second figure in chapter two as 2.2 and so on. All figures should be readable and clear. Captions should be consistently aligned (preferably centered). Elaborate each table appropriately, with at least 3 lines of description.

Clause 5: **List of figures** shall be generated using MS Word Table of figures/table method.

Clause 6: All **Algorithms (if/any)** should be written using proper algorithmic notation and you should also elaborate and show the time complexity of algorithm.

Clause 7: All **Functional and Non-Functional requirements** shall be proper numbered e.g. FR1 for functional requirement 1 and NFR1 for non-functional requirement. Non- Functional requirements are constraints.

Clause 8: A **Use case** should cover at least one or more transactions and should belong to some requirement. Every Use case should belongs to some functional or non-functional requirement.

Clause 9: Every **user behavioral use case** should has some associated actor. A use case without actor shall be directly linked with other use case using <<extends>>, <<include>> and <<Generalized> stereo type relationship.

Clause 10: Every **actor** should has some associated use case, two or more actors can only be directly linked with each other with <<Generalized> stereo type relationship.

Clause 11: Use **actor generalization** for common use cases. For example, Login, Logout, change Password are general use cases and should be linked with some General actor.

Clause 12: A **use case elaboration** shows proper dialog between actor and system in main flow and all exception flow references should be mentioned in main flow and explained in exceptional flow section.

Clause 13: There are two types of **Sequence Diagrams**:

- a. System Sequence Diagram
- b. Object Sequence Diagram

Clause 14: A use case can have **one or more system sequence diagrams** depending on the number of flows. Only two objects should be shown in system sequence diagram i.e. Actor as an entity object and system as a business object. Message passing between objects should be synchronized with use case elaboration.

Clause 15: A use case can have **one or more object sequence diagrams**. System object shown in the system should be decomposed into objects that will be used in Object sequence diagram. Objects shown in sequence diagram should cover the responsibilities given in use case elaboration. Message passing, life time of objects and time sequence should be clearly visible as flow in sequence diagram.

Clause 16: **Main** and **exception flows** should be covered in different Object sequence diagrams.

Clause 17: A Flattened relationship without time line can be shown in **Collaboration Diagram**.

Clause 18: An **activity diagram** should show responsibilities of objects with proper information and message flow.

Clause 19: Every dynamic object should have **state chart diagrams** with one start and more than one ending states.

Clause 20: For static (semantic) modeling, **class diagrams** are used for system design. Class diagram should conform to UML specification. Classes should not be isolated in system. Every class should be linked with each other using proper relationship, i.e. Association, Aggregation, Composition, Generalization, Inheritance and realization.

- i) Two classes linked with association should have association name and multiplicity.

- ii) Two classes linked with aggregation should show loosely coupled (Hollow Diamond) relationship with multiplicity.
- iii) Two classes linked with Composition should show loosely coupled (Filled black Diamond) relationship with multiplicity.
- iv) Two classes linked with Generalization should show **IS-A** relationship using some proper hierarchy and discriminant.
- v) Two classes linked with Inheritance should show **IS-A** relationship (using some proper hierarchy) of parent and child classes. Child class should extend functionality of parent class in terms of data and operations.
- vi) Two classes linked with Realization should show **IS-A** relationship using some proper hierarchy of abstract and concrete derived classes. Concrete derived classes should implement the operations given in abstract class.
- vii) Interfaces can be shown in realization relationship.
- viii) Late binding and polymorphism concept should also be shown using realization diagram.

Clause 21: Database Design should be shown using following five artifacts:

- i) ER Model
- ii) Relational Model
- iii) Verify your Relational model using Normalization (i.e. Lossless and Dependency Preserved)
- iv) Data dictionary
- v) Table Script (SQL DDL)

Clause 22: An ER diagram should follow some notation. Following are some popular notations used in first database course by top 50 QS ranking universities:

- i) *Korth & Silberschatz (Silberschatz, 2010)*
- ii) *Elmasri & Navathe (R. Elmasri, 2011)*
- iii) *Chen (Peter-Pin-Shan, 1976), (Chen", 1983)*
- iv) *Bachman (Bachman, 1992)*
- v) *Batini, Ceri , and Navathe (C. Batini, 1992)*
- vi) *Information Engineering (KnowledgeWare, 1991), (IEF, 1990), (Martin, 1990)*
- vii) *Teorey (Teorey, 1990), (Teorey, 1999)*
- viii) *Oracle's CASE*METHOD (Barker, 1990)*
- ix) *IDEF1X Information Mode (T.Bruce, 1992)*

You should mention which notation you choose for your ER model.

Clause 23: Attach actor wise **screenshot GUI flow**.

Clause 24: Every Screenshot (given in GUI Flow) from your executable system should be captioned, explained and referred in the text.

Clause 25: Screenshots should be explained in two parts:

- a. Screen Description
- b. User Actions

Clause 26: Provide **references** in literature review/similar products section. Follow a citation style consistently. Provide bibliography of all the references cited in the text.

Clause 27: **Spacing** should be proper and uniform throughout the document. No white spaces should be there (do not leave half of the page blank after/before figures or use-cases).

Clause 28: There should not be “**sir**” written in front of advisor's name. It will be a well know salutation like Mr./Ms./Mrs./Dr./Engr./ Prof. Dr.

Clause 29: Use **one type of font**, **one size for headings** and **one size for subheadings** throughout the documentation. In other words, formatting should be consistent throughout the text.

Clause 30: **Bullets** at one level should be of same type.

Clause 31: **Text placeholders** in the FYP documentation template are surrounded with angular brackets, e.g. <project name>. Remove angular brackets when the placeholder is replaced with actual text.
